

Data Modeling in Power BI

Data Modeling in Power BI is the process of **organizing and structuring your data** so that reports and dashboards can be created efficiently. It ensures that **tables relate correctly**, and calculations are accurate.

Key Concepts in Power BI Data Modeling

1. Tables

- Imported datasets (Excel, SQL, CSV, etc.)
- Can be fact tables or dimension tables (explained below)

2. Relationships

- Define how tables are connected.
- Types:
 - **One-to-Many (1:N)** → Most common
 - **Many-to-Many (N:N)** → Rare, use with care
- Use primary key → foreign key mapping

3. Calculated Columns

- New columns derived from existing columns using DAX formulas.
- Example: **Profit = Revenue - Cost**

4. Measures

- Aggregations computed at query time using DAX.
- Example: **Total Sales = SUM(Sales[Amount])**

5. Hierarchies

- Organize data in levels for drilling down.
- Example: **Year → Quarter → Month → Day**

6. Star & Snowflake Schema

- Structuring tables for efficient analysis.



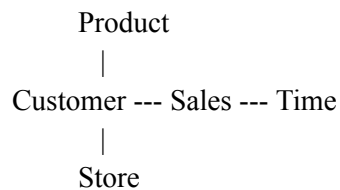
Star Schema

Star Schema is a simple data modeling approach where a **central fact table** connects directly to **dimension tables**.

Characteristics

- Fact table in the center
- Dimension tables around it (like a star)
- Denormalized dimension tables (i.e., fewer joins)
- Optimized for **query speed and reporting**

Structure



Example

- **Fact Table:** **Sales**
 - Columns: **SalesID**, **ProductID**, **CustomerID**, **StoreID**, **Quantity**, **Revenue**
- **Dimension Tables:**
 - **Product** → ProductID, Name, Category
 - **Customer** → CustomerID, Name, Region
 - **Store** → StoreID, Name, Location
 - **Time** → Date, Month, Quarter, Year

✓ Pros:

- Simple to understand
- Faster queries

- Easier for reporting and Power BI visuals

✗ Cons:

- Dimension tables can be large (denormalized)
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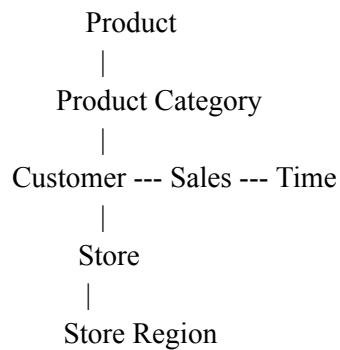
3 Snowflake Schema

Snowflake Schema is a more **normalized version** of the star schema. Dimension tables are **split into multiple related tables**.

Characteristics

- Fact table in the center
- Dimension tables may have **sub-dimensions**
- More joins required
- Optimized for **storage efficiency**

Structure



Example

- **Fact Table:** Sales
 - Same as star schema
- **Dimension Tables:**
 - **Product** → ProductID, Name, CategoryID
 - **ProductCategory** → CategoryID, CategoryName
 - **Customer** → CustomerID, Name, RegionID
 - **Region** → RegionID, RegionName

✓ Pros:

- Reduces data redundancy

- Saves storage

✗ Cons:

- More complex
 - Slower query performance due to multiple joins
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